Movie Explorer 2.0

Build a single-page React application that lets users **search movies by title**, browse a **poster grid** of results from **TMDB**, and maintain a personal **Favourites** ribbon that persists across page reloads. This sprint focuses on real-world async flows (`fetch + async/await`), React lifecycle with `useEffect`, local persistence, UI states (loading / error / empty), and a clean visual polish (hoverable cards, grey poster placeholders, sticky favourites bar).

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1 · Learning Objectives

Concept / Skill	Practised Through
Controlled components & debouncing	`<SearchBar>` input → `query` state
Custom hooks	`useMovieSearch(query)` (data, loading, error)
Side-effects & cleanup	`useEffect` 🏷 on mount + on `query` change
Immutable state patterns	Adding / removing favourites
LocalStorage hydration/persistence	Two dedicated `useEffect` hooks
Conditional rendering & skeletons	Spinner, error banner, “No results”
Responsive grid & subtle animations	CSS Grid + hover transforms + shadow
Environment variables	`VITE_TMDB_API_KEY` via `.env`
Accessibility basics	`aria-labels` on interactive icons

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2 · Setup & Running

1. **Clone / fork** the starter repo (or create a Vite + React project).
2. Generate a **TMDB account & API key** (free):
   - Sign-up → Profile → **Settings › API** → request **v3 key** (32-char).
3. Add a `.env` file at project root *(never commit the key)*:

   ```
   VITE_TMDB_API_KEY=xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
   ```

4. Place `placeholder.svg` (200×300 grey poster) inside `public/`.
5. `npm i` → `npm run dev` → open http://localhost:5173.

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3 · Functional Requirements (“Definition of Done”)

ID	Feature	Acceptance Criteria

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F-1	**Search** bar	Typing updates `query` after 400 ms; empty query clears grid.
F-2	**Loading spinner**	Visible while awaiting TMDB response; disappears on fulfillment / error.
F-3	**Error banner**	Red message if network fails or TMDB returns no matches.
F-4	**Poster grid**	Up to 10 cards: poster (or placeholder), title, year, star icon.
F-5	**Favourite toggle**	Star adds/removes movie from favourites in immutable way.
F-6	**Sticky favourites bar**	Shows poster thumbnails; ❌ icon removes entry.
F-7	**Persistence**	Favourites survive hard reload via `localStorage`.
F-8	**Responsive layout**	Cards re-flow; hover lift + shadow; placeholder nicely fills missing posters.
F-9	**Accessibility**	Every button focusable; `aria-label` explains action.
F-10	**Zero console warnings**	ESLint clean; no “act” warnings or unhandled rejections.

4 · Stretch Goals (optional +10 pts)

1. **Infinite scroll / “Load more”** pagination.
2. **Modal details panel** (second TMDB call by movie ID).
3. **Dark-mode toggle** with persisted preference.
4. **Custom `usePersistentState` hook** to generalise LS logic.
5. **Keyboard grid navigation** (arrow keys + Enter toggles favourite).